New Science

Revise this chapter together

Unit 1 About the Earth

3 Weather and Climate





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1. The combined conditions of the atmosphere in a place over a short period of time is called weather. The average weather condition in a place over a longer period of time is called climate.

Visit the Hong Kong Observatory website. Look at how the Hong Kong Observatory describes **the weather** and fill in the information about the weather today.



Quick Check

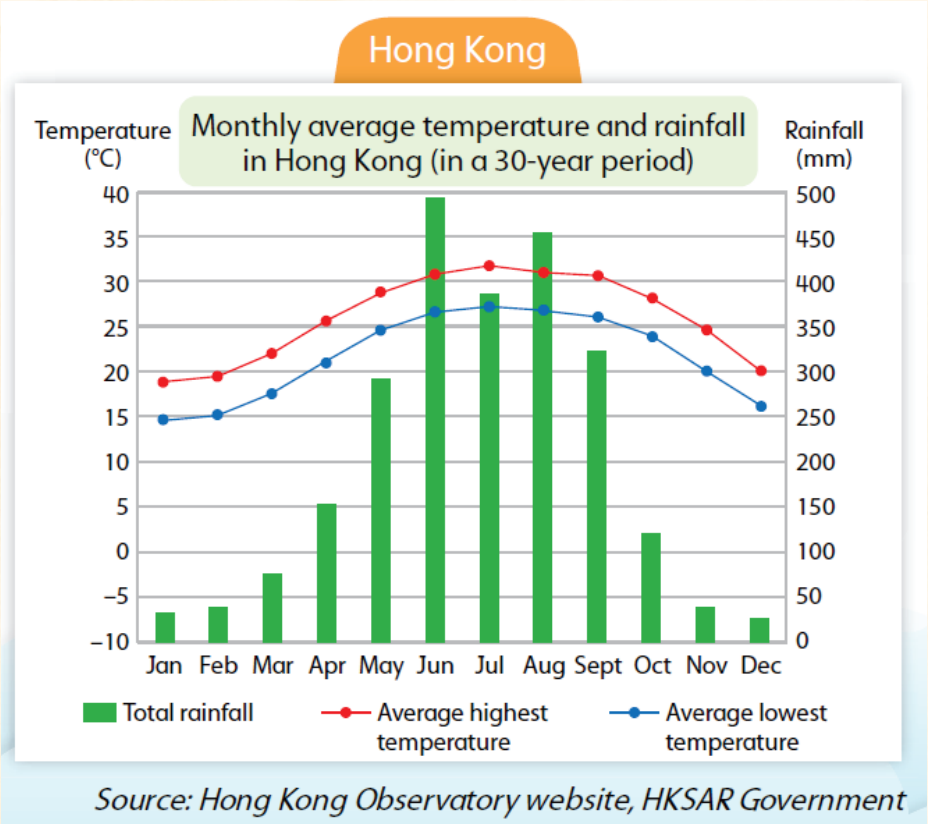
The Hong Kong Observatory:

1. uses temperature / relative humidity to describe the coldness and the hotness of the weather.
2. uses temperature / relative humidity to describe the dryness and wetness of the weather.
3. uses a picture to show whether there is rainfall.

1. The combined conditions of the atmosphere in a place over a short period of time is called weather. The average weather condition in a place over a longer period of time is called climate.

Is Hong Kong's temperature today **normal**? Find out the answer from this graph?

Weather changes every day. However, when we observe it for a long period of time, we can see that there is a regular _pattern_ for the changes.

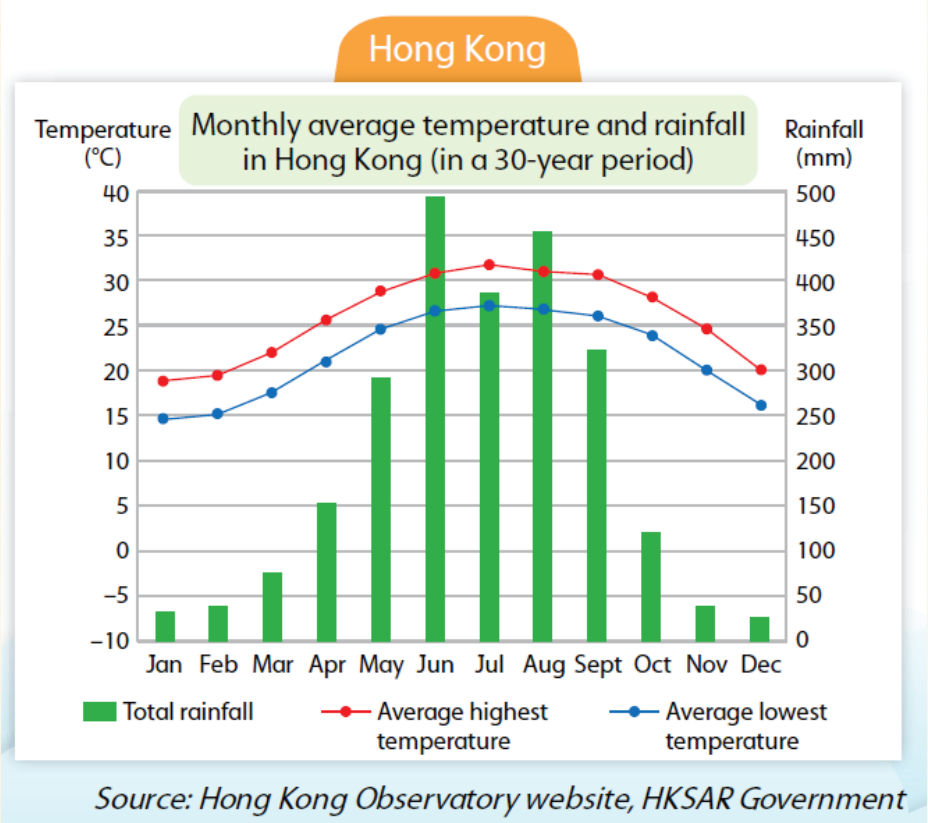


Source: Hong Kong Observatory website, HKSAR Government

1. The combined conditions of the atmosphere in a place over a short period of time is called weather. The average weather condition in a place over a longer period of time is called climate.

Is Hong Kong's temperature today **normal**? Find out the answer from this graph?

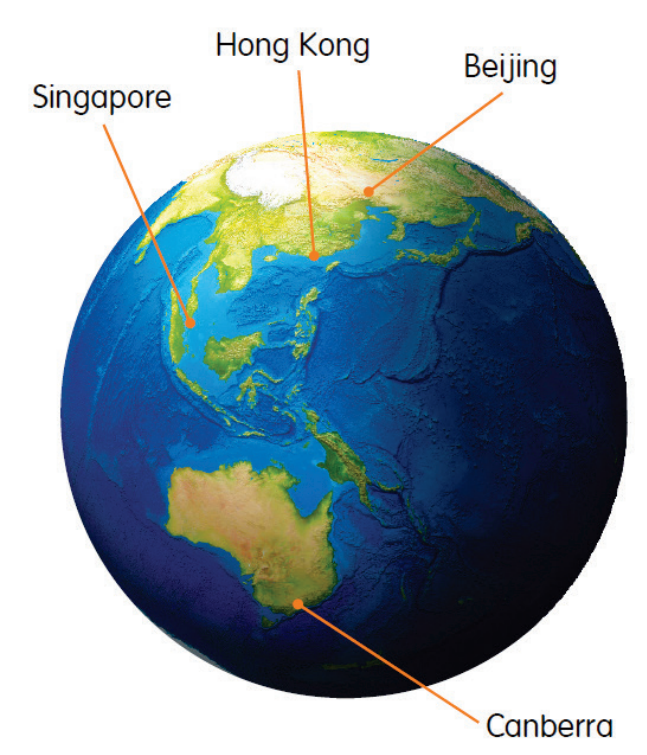
Such a pattern is referred to as _____climate_____. It is the _____average_____ weather condition over a longer period of time in the past.



Source: Hong Kong Observatory website, HKSAR Government

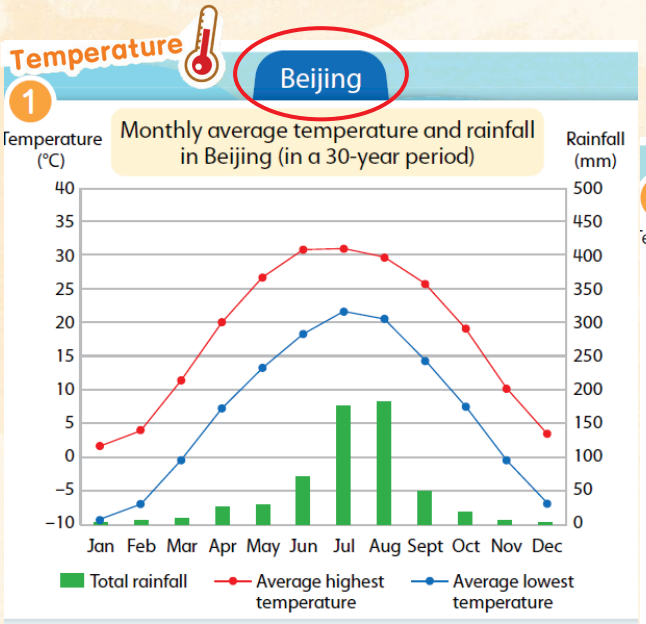
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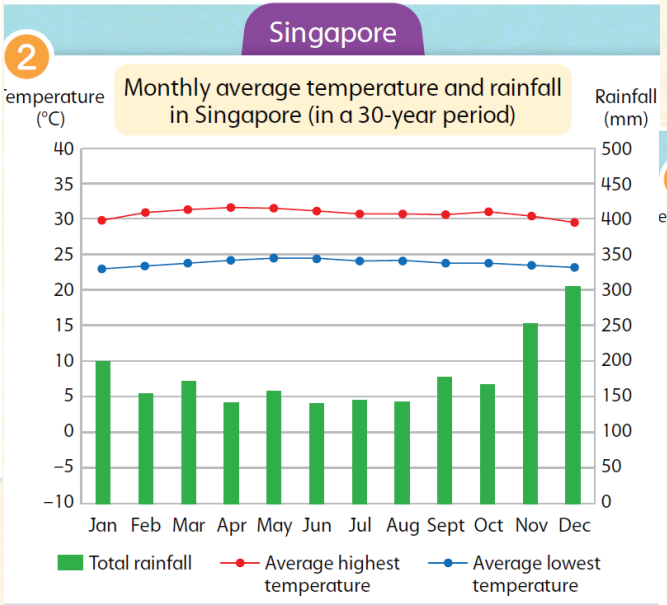
Compare the temperature, rainfall and relative humidity in Hong Kong, Beijing, Singapore and Canberra. Find out their **climate characteristics** and answer the questions.

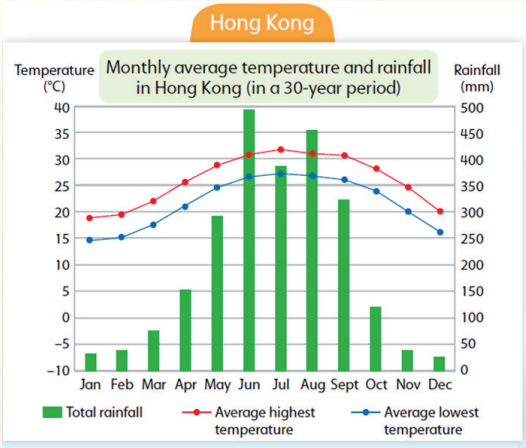


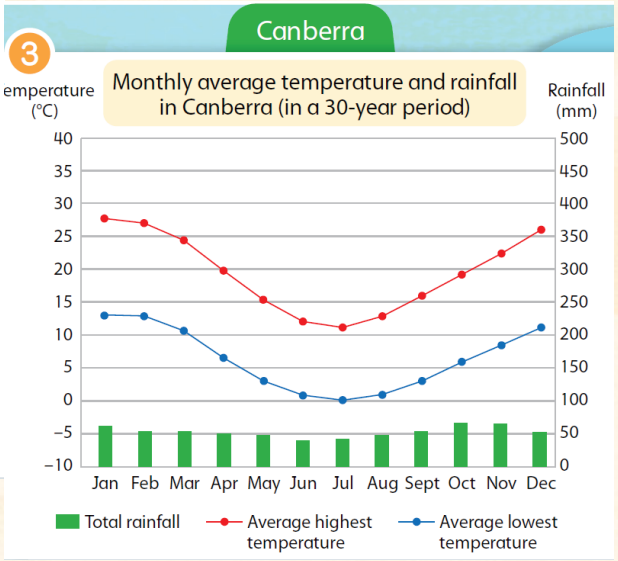
2. The climate of different places may be different because of their geographical locations.

Which place is **similar** to Hong Kong in terms of **temperature changes over the year**?



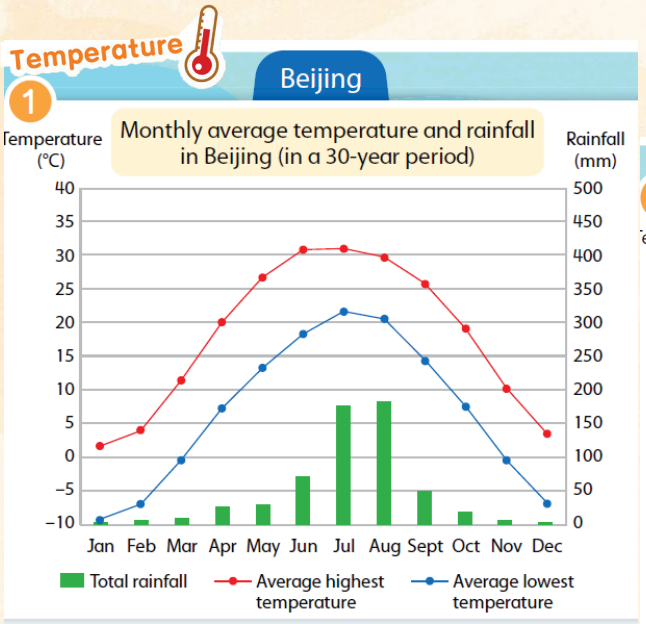


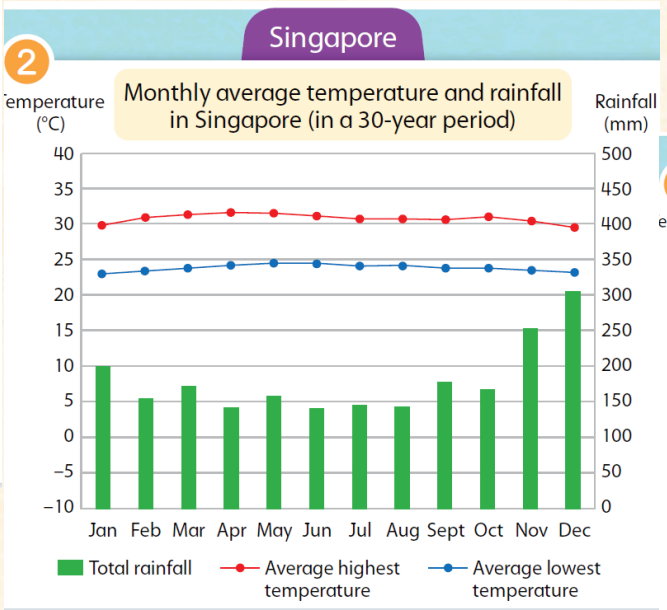


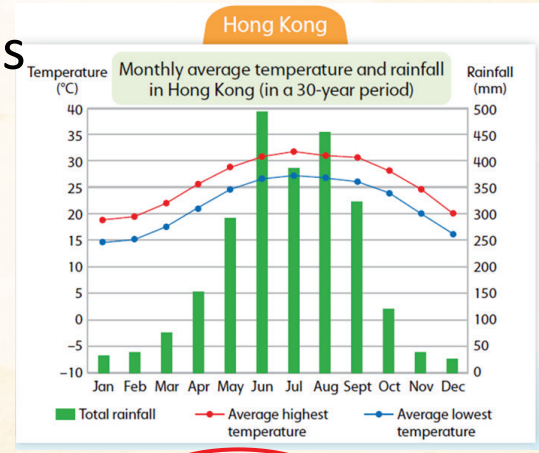


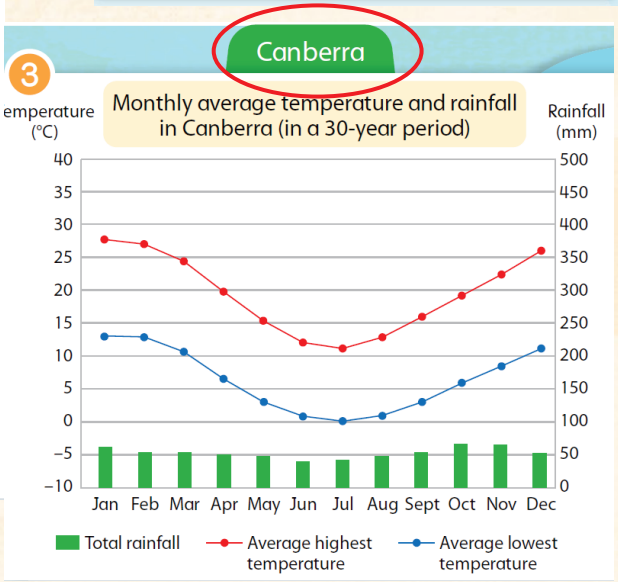
2. The climate of different places may be different because of their geographical locations.

Which place is **opposite** to Hong Kong in terms of **temperature changes over the year**?



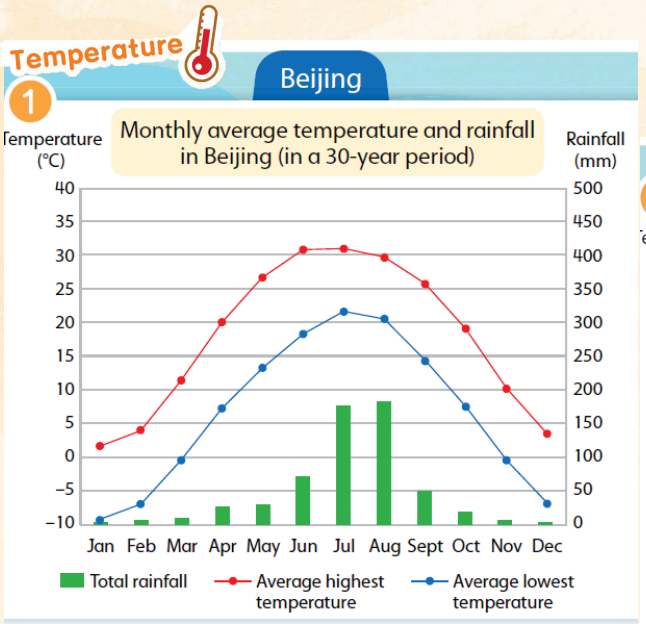


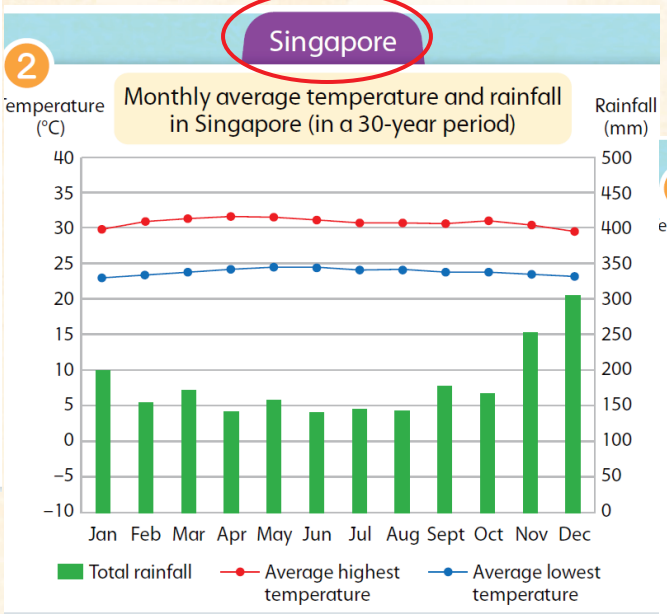


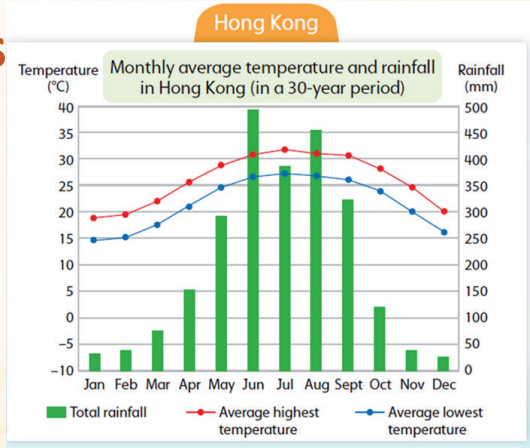


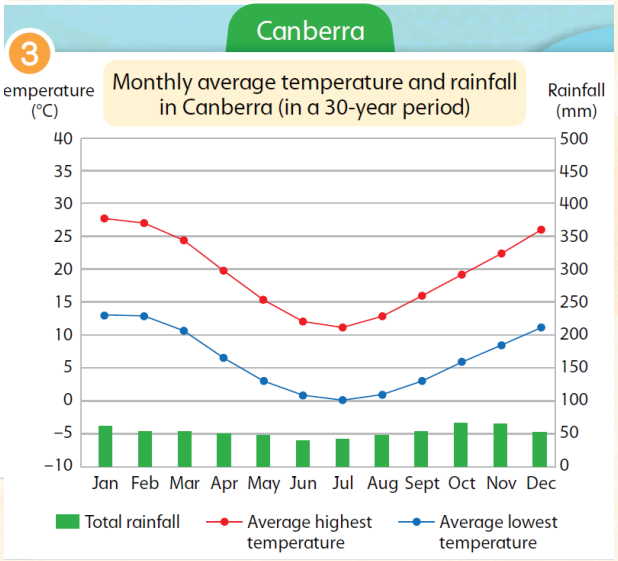
2. The climate of different places may be different because of their geographical locations.

Which place has **smaller temperature changes over the year**?





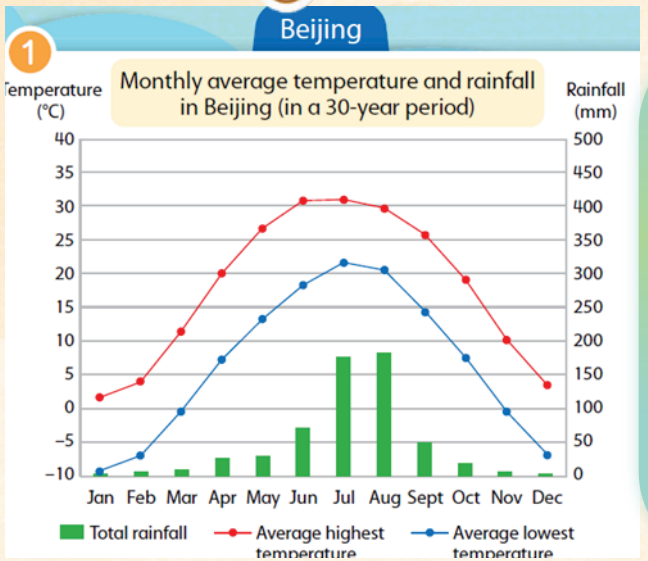




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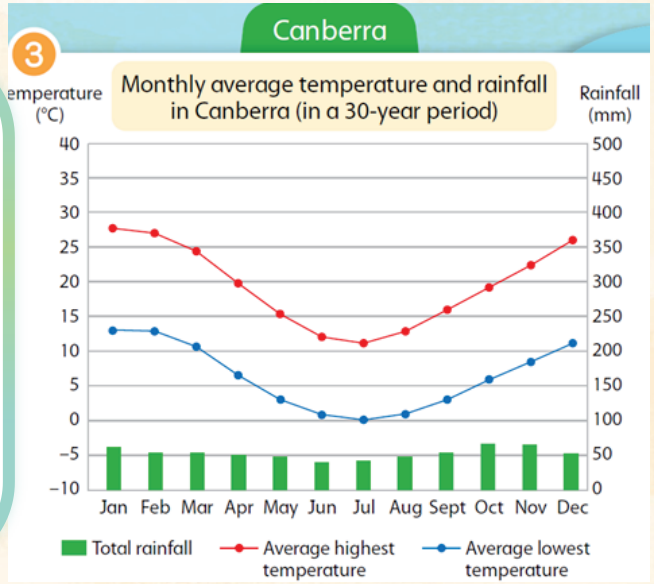
How are the **seasons** of these places related to their **locations** on the Earth?

Temperature



Beijing is in the northern hemisphere / southern hemisphere .

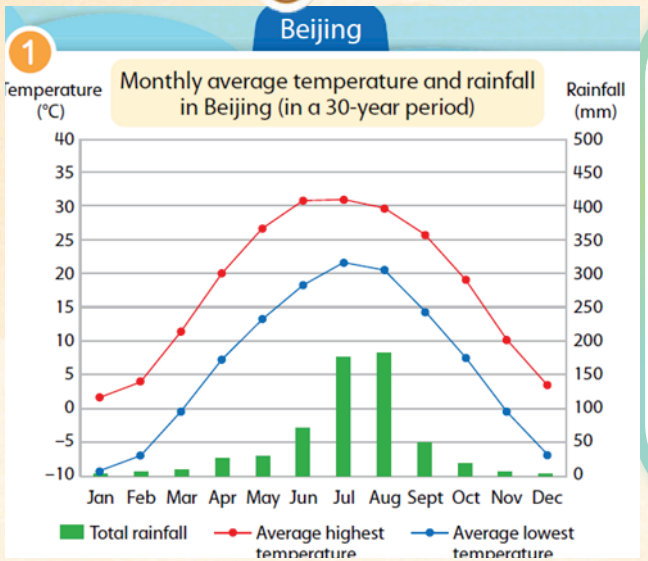
When it is winter in Beijing, it is winter / summer in Canberra.



2. The climate of different places may be different because of their geographical locations.

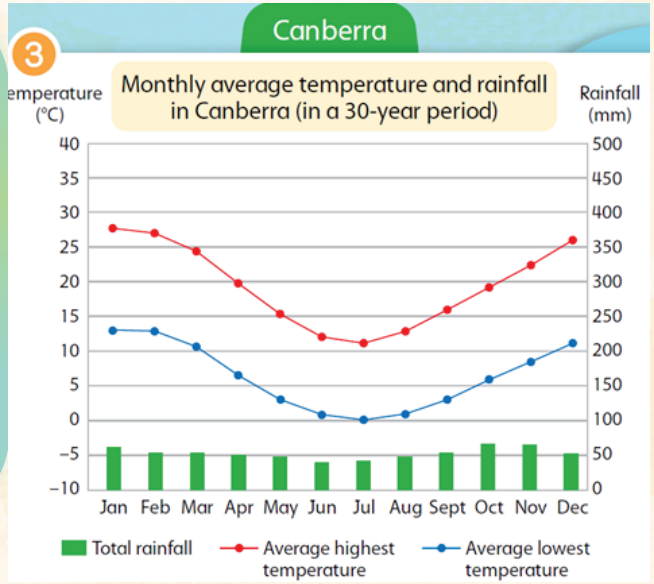
How are the **seasons** of these places related to their **locations** on the Earth?

Temperature



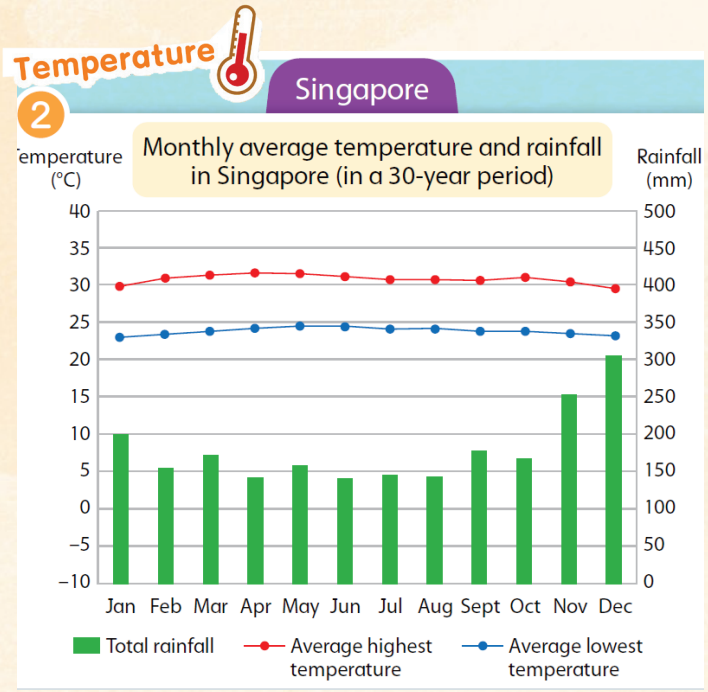
Canberra is in the northern hemisphere / southern hemisphere .

When it is summer in Beijing, it is winter / summer in Canberra.



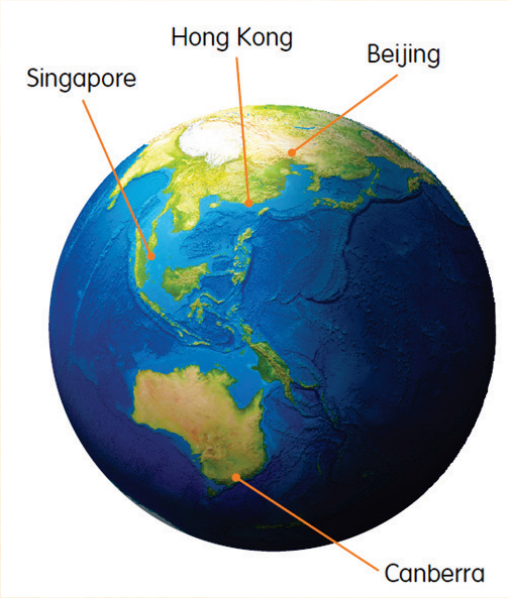
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How are the **seasons** of these places related to their **locations** on the Earth?



Singapore is near / far from the equator.

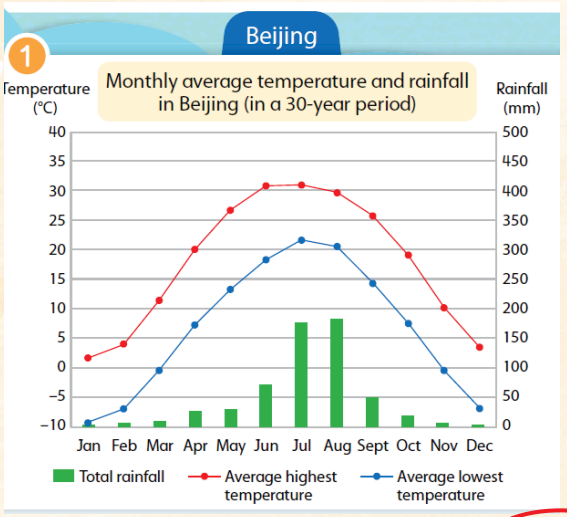
Singapore has / does not have four distinct seasons.

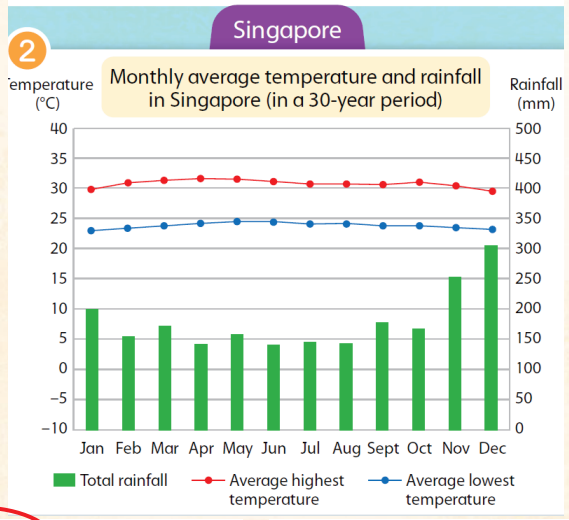


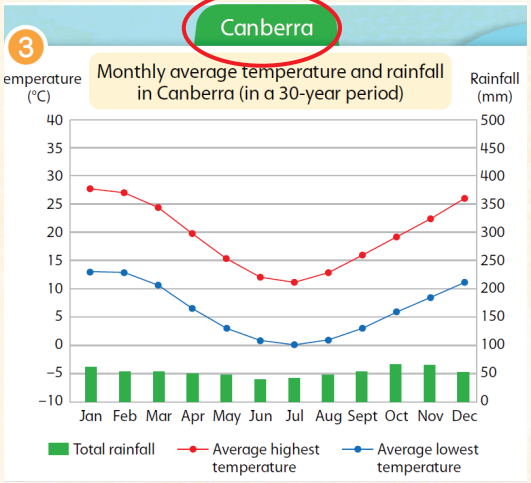
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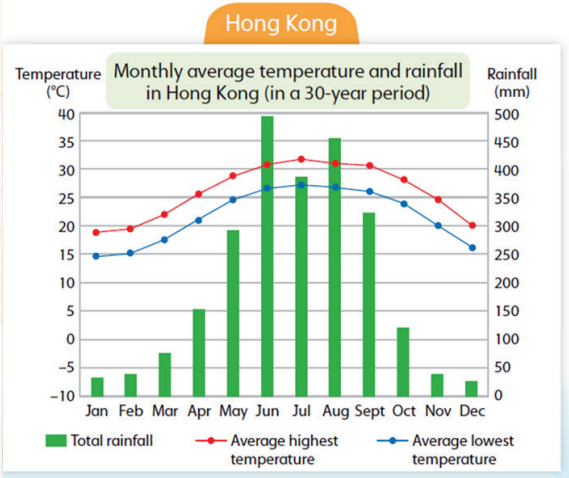
Which place has **the lowest rainfall** in July?

Rainfall





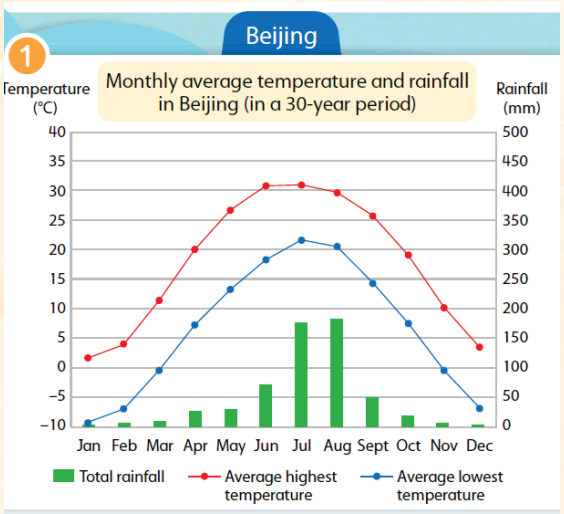


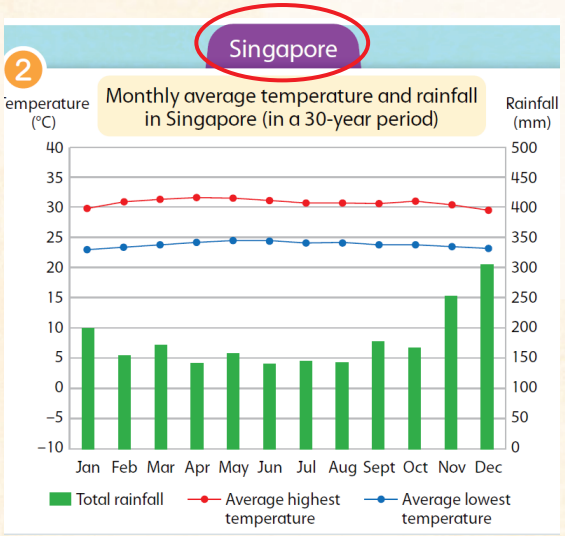


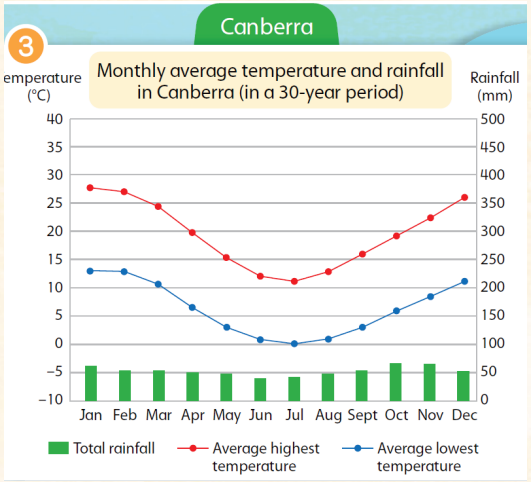
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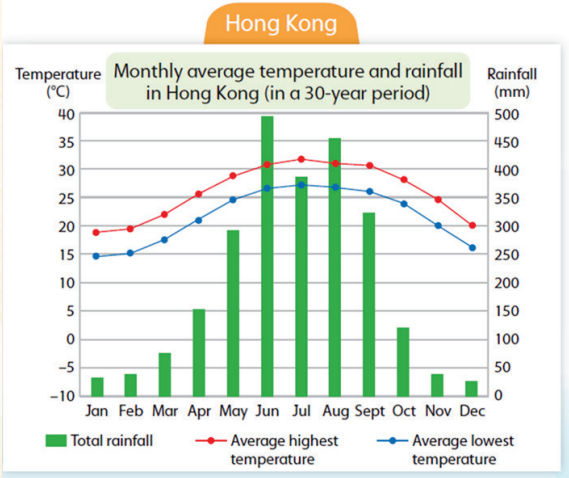
Which place has **high rainfall** over the year?

Rainfall









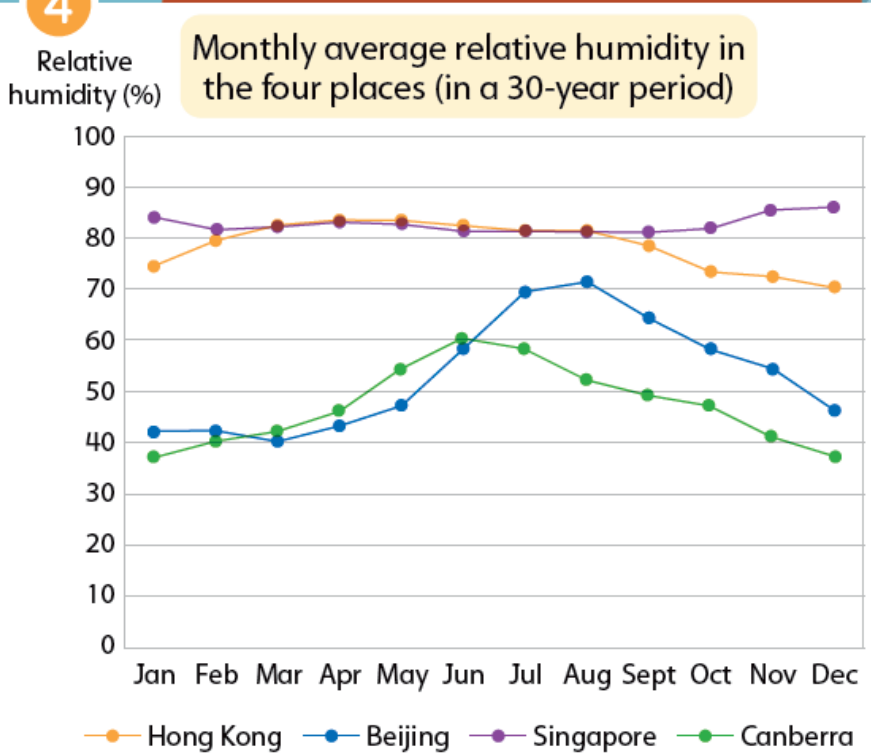
2. The climate of different places may be different because of their geographical locations.

Which places are **wetter**?

Relative humidity

Comparing the relative humidity of the four places

4



Hong Kong / Beijing / Singapore / Canberra

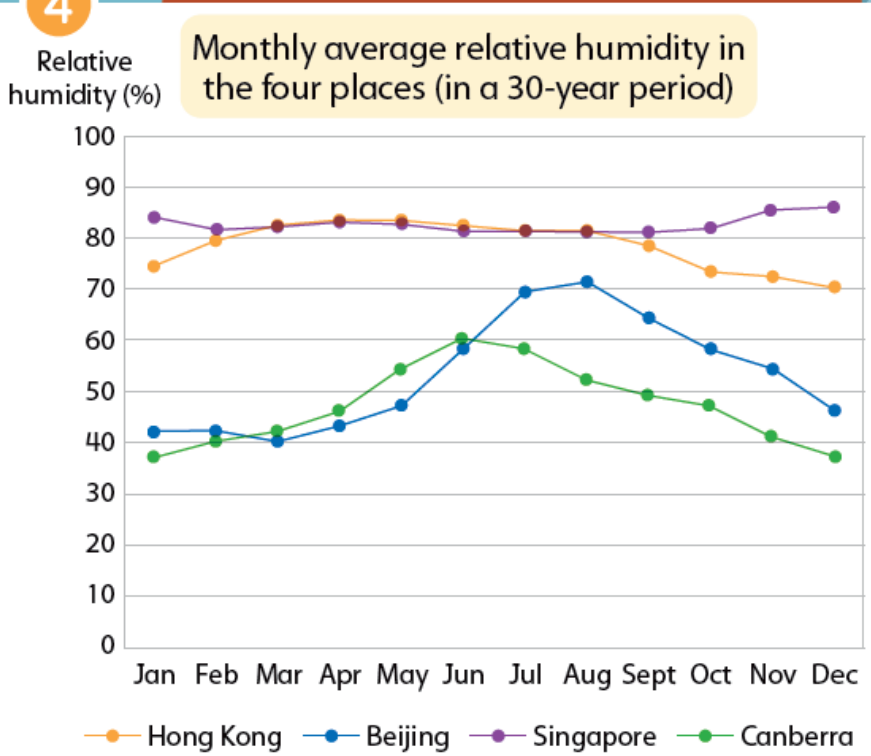
2. The climate of different places may be different because of their geographical locations.

Which places are **drier**?

Relative humidity

Comparing the relative humidity of the four places

4



Hong Kong / Beijing / Singapore / Canberra

2. The climate of different places may be different because of their geographical locations.

Jenny and her family are planning to have a ‘white Christmas’. Which of these places should they visit to have the best chance of seeing **snow**?

Hong Kong / Beijing / Singapore / Canberra



2. The climate of different places may be different because of their geographical locations.

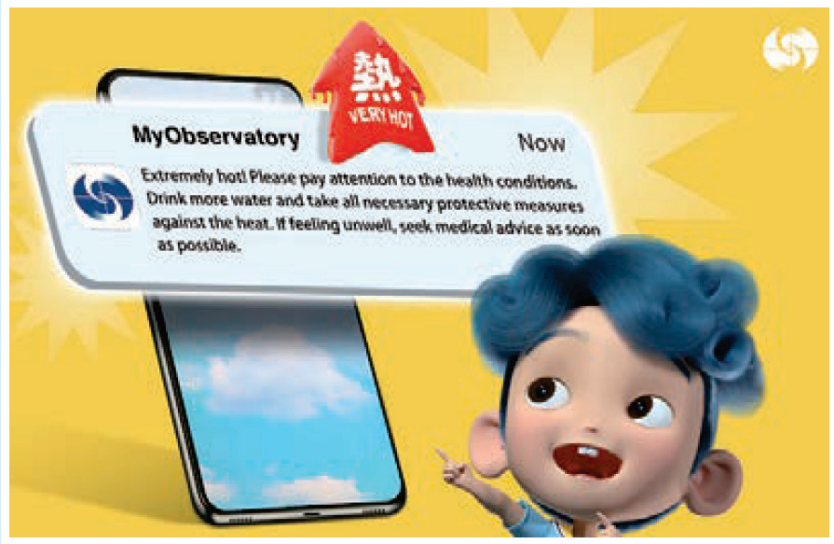
Work in groups. Among the four places, discuss which place you like the most in terms of its **climate**. Why?

Hong Kong / Beijing / Singapore / Canberra



3. When the temperature is 33°C or above, the Hong Kong Observatory will issue the Very Hot Weather Warning to remind people to stay hydrated and take measures to prevent heatstroke.

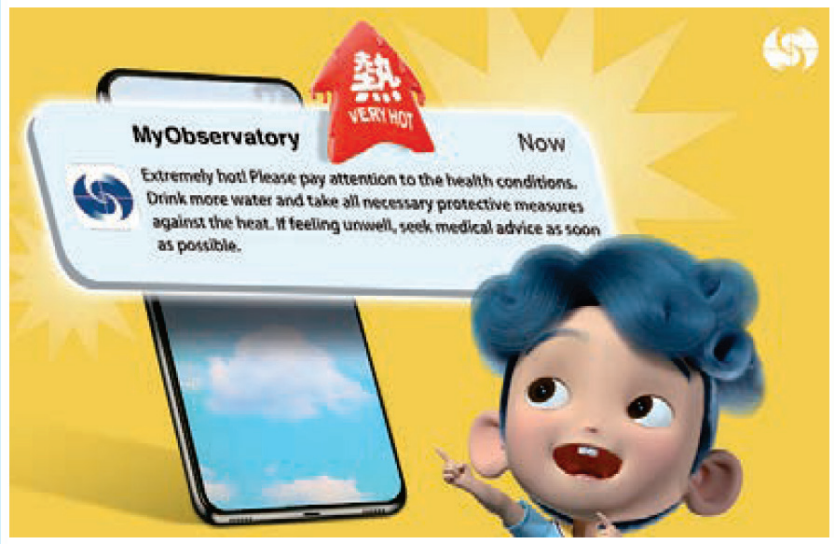
What does '**very hot weather**' mean?



The highest temperature reaches 33 °C or above.

3. When the temperature is 33°C or above, the Hong Kong Observatory will issue the Very Hot Weather Warning to remind people to stay hydrated and take measures to prevent heatstroke.

Why does the Hong Kong Observatory need to enhance the **Very Hot Weather Warning** service?



As _____ persists, Hong Kong has become hotter and hotter. Enhancing the Very Hot Weather Warning service can remind the public of taking protective measures against the heat.

3. When the temperature is 33°C or above, the Hong Kong Observatory will issue the Very Hot Weather Warning to remind people to stay hydrated and take measures to prevent heatstroke.

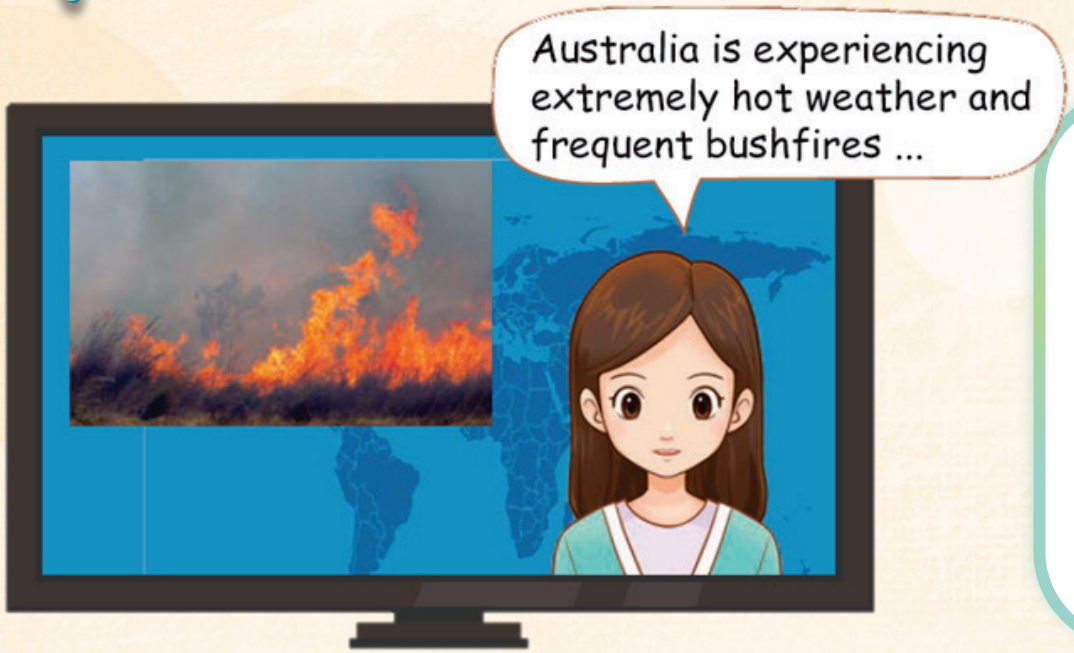
Do you feel that it is **getting hotter** in Hong Kong **every year**? Do you have scientific evidence for this feeling?



Read the weather data recorded by the Hong Kong Observatory to find out the evidence.

3. When the temperature is 33°C or above, the Hong Kong Observatory will issue the Very Hot Weather Warning to remind people to stay hydrated and take measures to prevent heatstroke.

Does the **rise of temperature** only happen in Hong Kong?

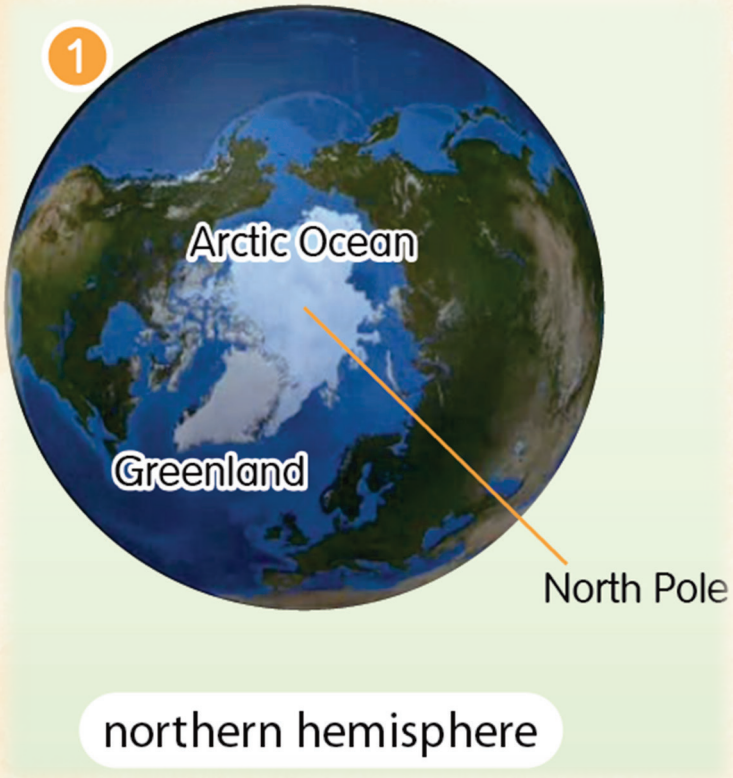


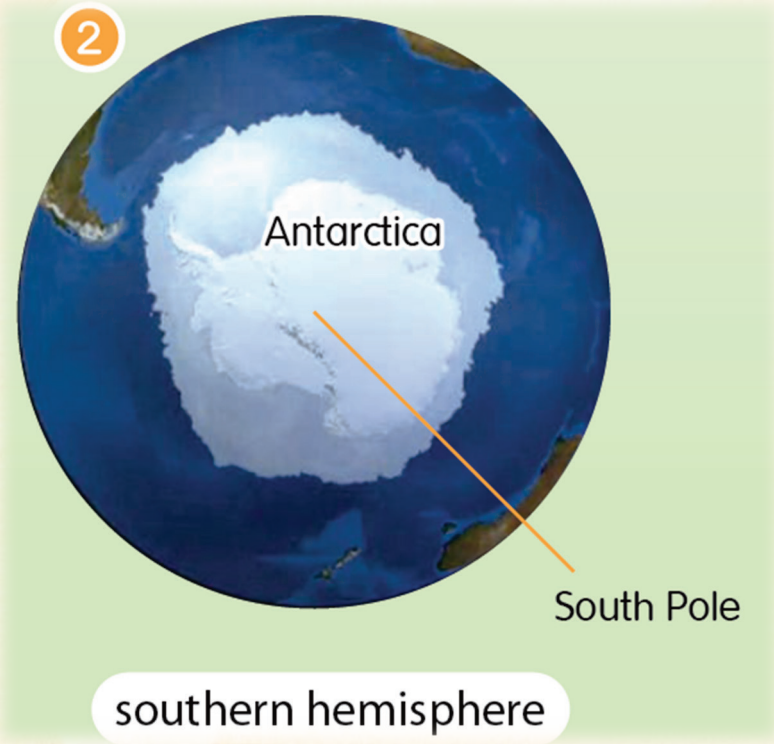
Yes / No , e.g. the rise of temperature also happens in Australia . The extremely hot weather there caused bushfires.

4. Global warming leads to environmental problems, such as melting land ice, rising sea levels, and desertification.

What place on the Earth is **covered with snow and ice** all year around?

The North Pole and the South Pole.

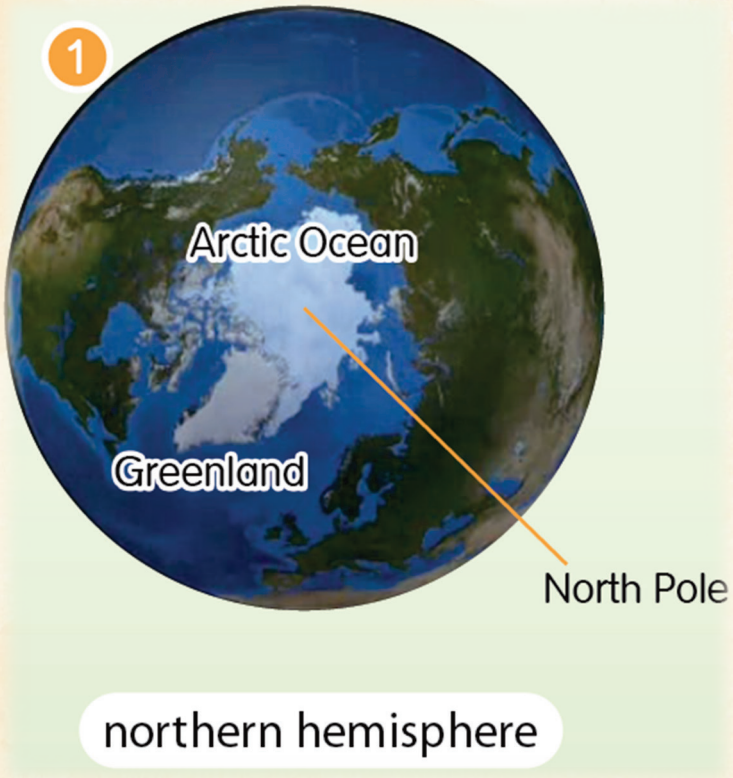


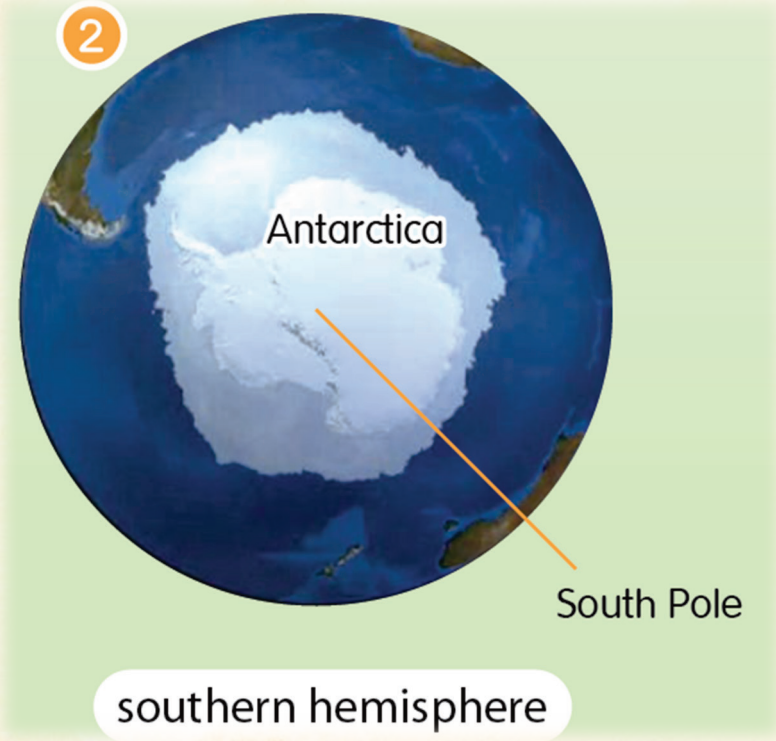


4. Global warming leads to environmental problems, such as melting land ice, rising sea levels, and desertification.

There is a lot of snow and ice in the oceans and on the land of these places. How are these places **affected** by global warming?

The ice and snow in these places will melt.





4. Global warming leads to environmental problems, such as melting land ice, rising sea levels, and desertification.

Simulate the **effect** of melting sea ice and land ice **on sea levels**.

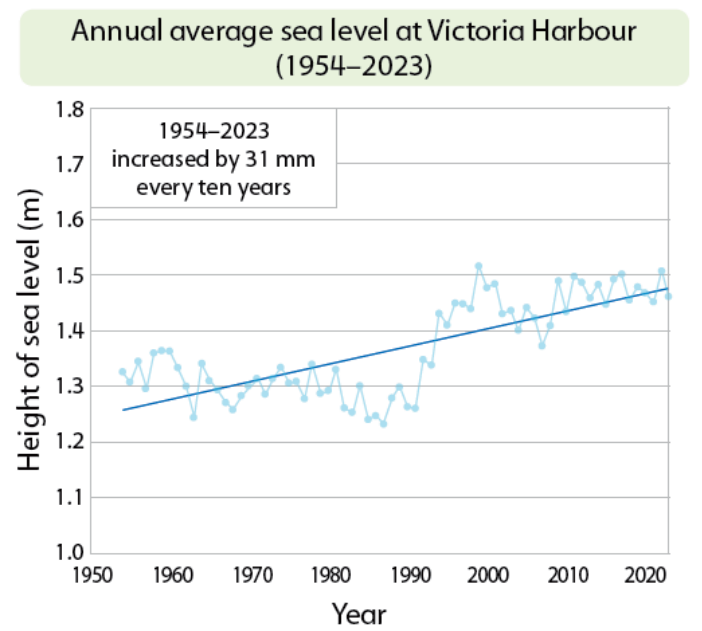




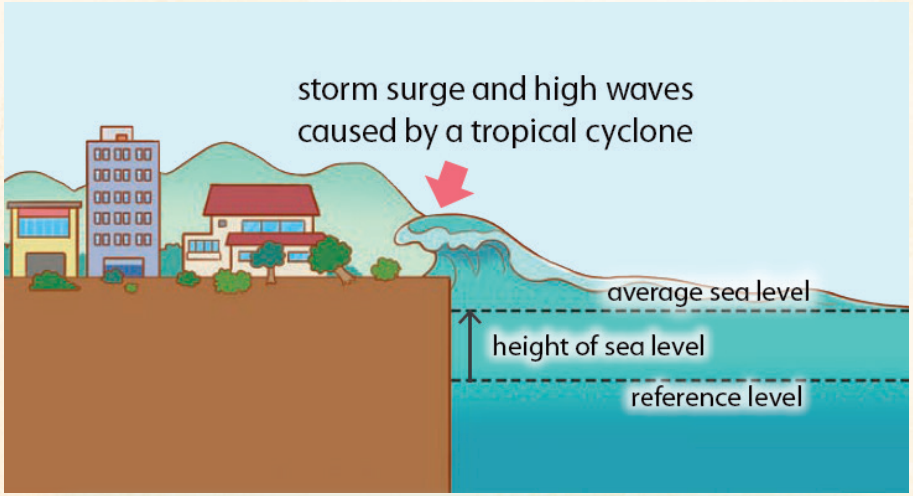
Melting sea ice / land ice causes sea levels to rise.

4. Global warming leads to environmental problems, such as melting land ice, rising sea levels, and desertification.

How may the rise of sea levels **affect us**?



Source: Hong Kong Observatory website, HKSAR Government



It may increase the threats of storm surge and high waves when a tropical cyclone hits. The coastal areas may be flooded with sea water.

4. Global warming leads to environmental problems, such as melting land ice, rising sea levels, and desertification.

What other **effects** does global warming bring?





The rising temperature changes the distribution of rain. This may worsen the _drought_ in some places, causing _desertification_, while other places may be affected by _floods_.

4. Global warming leads to environmental problems, such as melting land ice, rising sea levels, and desertification.

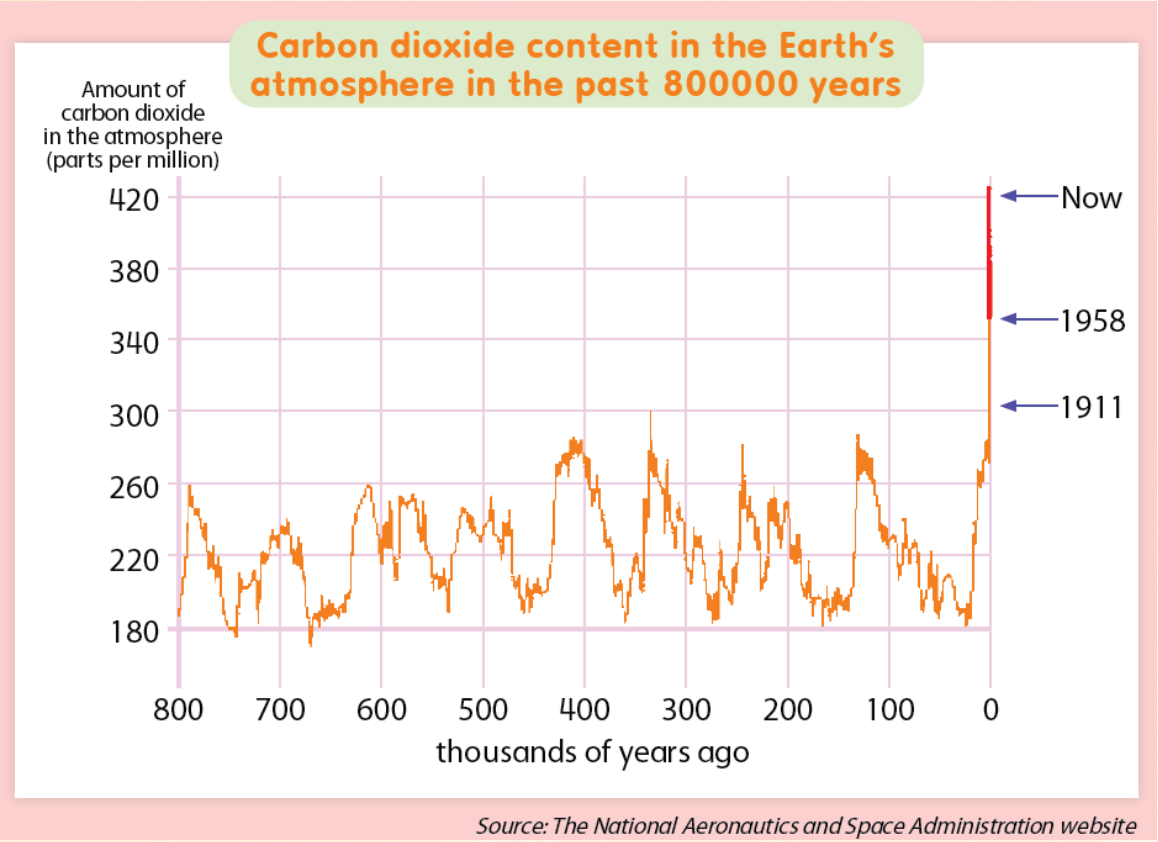
What other **effects** does global warming bring?



When there is no vegetation covering the land, the soil and dust will be exposed, and sandstorms will occur easily. These situations not only affect the survival of animals and plants, and our food supply, they may even cause causalities.

5. Global warming happens because humans use fuels, such as coal and oil, excessively, releasing too many greenhouse gases.

In the past one hundred years, how has **the amount of carbon dioxide** in the Earth's atmosphere changed?



Source: The National Aeronautics and Space Administration website

The amount of carbon dioxide has increased / decreased sharply.

5. Global warming happens because humans use fuels, such as coal and oil, excessively, releasing too many greenhouse gases.

Why has the amount of carbon dioxide in the Earth's atmosphere **increased** sharply in the past one hundred years?









Human activities nowadays, such as _transportation_ and industrial activities, and some natural phenomena, such as _bushfires_, all release carbon dioxide. Besides, frequent deforestation reduces trees that can absorb carbon dioxide, thus increases the amount of carbon dioxide in the atmosphere greatly.

5. Global warming happens because humans use fuels, such as coal and oil, excessively, releasing too many greenhouse gases.

What **impacts** do increasing greenhouse gases have on the Earth's surface?



Large amounts of greenhouse gases block the _heat_ from releasing from the Earth's surface and cause a _rise_ in the Earth's temperature.

6. Carbon footprints are the amount of greenhouse gases produced in some human activities.

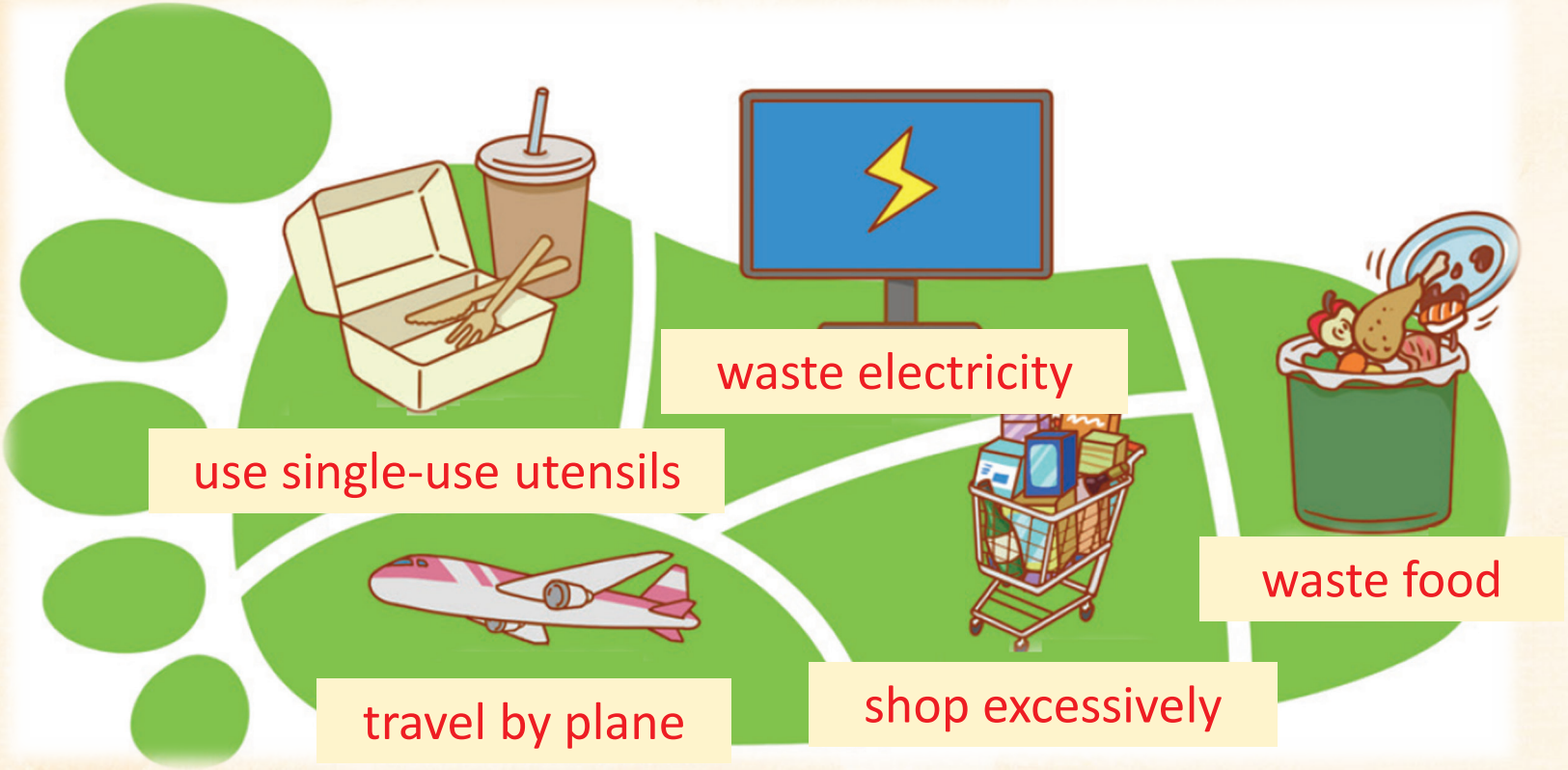


What is a **carbon footprint**?

Carbon footprints are the amount of greenhouse gases produced in some human activities. Many daily activities leave carbon footprints.

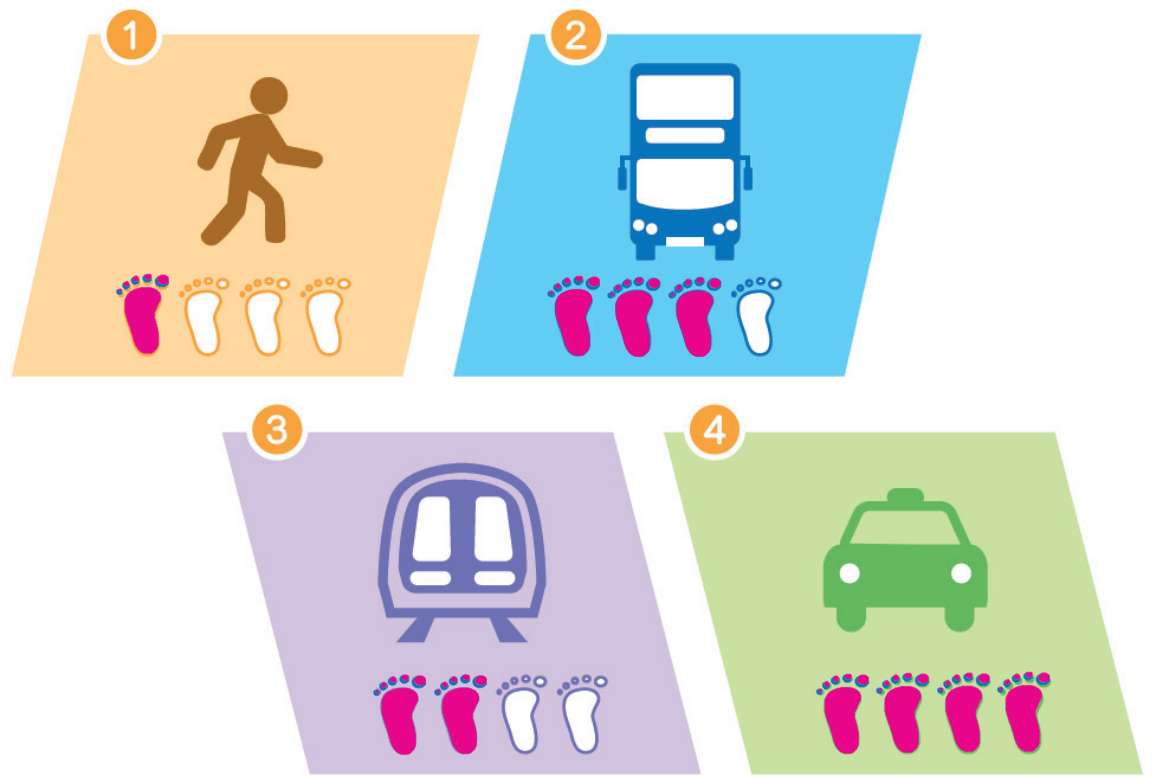
6. Carbon footprints are the amount of greenhouse gases produced in some human activities.

What activities or behaviour **leave carbon footprints** unnecessarily?



7. If we reduce the emission of greenhouse gases, such as carbon dioxide and carbon footprints, we can slow down global warming.

Which of these means of transport leaves fewer personal **carbon footprints**? Which one leaves more? Rank them and colour the footprints. (1= the fewest; 4 = the most)



7. If we reduce the emission of greenhouse gases, such as carbon dioxide and carbon footprints, we can slow down global warming.

Choose a topic on **climate change** (e.g. the causes of global warming, ways to slow down global warming and carbon footprints). Design a poster and share the information with your classmates.







I have learned

Weather and climate of the Earth

Weather

condition of the atmosphere over a short period of time



Climate

average weather condition over a longer period of time

vary in different locations

Climate change

Global warming

Causes

excessive use of coal, oil, etc.

Effects

melting glaciers
rising sea levels
desertification

Ways to slow it down

reduce carbon footprints